

Roof Lifecycle Cost Analysis

San Antonio market
October 2, 2026

Planned maintenance is projected to save the owner of a typical San Antonio school **\$1.83M** over the next 20 years.

That is 37.6% less than fixing problems only after they leak. Under conservative to upside assumptions, the savings range from \$999K to \$2.36M. In today's dollars the estimate is worth \$949K.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE	NEW ROOF TODAY
San Antonio	K-12	120,000 sq ft	Mod bit	8 yrs Good	\$1.23M

PAXSEAL NET COST \$3.03M \$2.91M spent plus roof wear, net of roof life left	REACTIVE NET COST \$4.87M \$4.29M spent plus roof wear, net of roof life left	EMERGENCY COSTS CUT \$2.08M Unbudgeted repairs and storm damage avoided	NEW ROOF PRICE 2× in 15 yrs \$1.23M today, \$3.02M by year 20
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Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal program fees	\$441K	—	-\$441K
Repairs paid by owner	\$300K	\$2.3M	+\$2.0M
Expected storm damage	\$154K	\$231K	+\$77K
Roof replacements	\$2.02M	\$1.76M	-\$255K
Current roof value used	\$738K	\$738K	—
Credit for roof life left	-\$615K	-\$164K	+\$451K
Net 20-year cost	\$3.03M	\$4.87M	+\$1.83M

What this means

- 1 A later replacement. The next new roof moves from year 8 to year 11, when it costs less in today's dollars.
- 2 Smaller repairs. Fixes made on scheduled visits avoid emergency work at about 3.9× the cost in a school, saving \$2.0M.
- 3 A predictable budget line. Emergency repairs become a fixed annual cost the board can plan for, and leaks stop closing classrooms.
- 4 The program pays for itself. Each \$1 of fees avoids about \$5.15 of other roof costs.

Local risk profile

Bexar County, TX (48029)

Hail and tornado

Baseball-size hail somewhere in Bexar County about once every 3 years (NOAA); April 2016 storm ≈ \$1.4B damage. Expected storm damage to this roof over 20 years is **\$154K** if maintained and **\$231K** if neglected. A documented maintenance history also helps when an insurer questions a claim.

Hail	Very High	■■■■■
Strong wind	High	■■■■■
Tornado	Very High	■■■■■
Hurricane	Moderate	■■■■■
Winter weather	High	■■■■■
Ice storm	Moderate	■■■■■
FEMA National Risk Index rating		

Key assumptions

The basis for each figure in this analysis.

ASSUMPTION	VALUE	BASIS
Roof life	20 / 15 yrs	Maintained / reactive. Fannie Mae service-life tables; +5 yrs per NRC Canada and warranty terms
Roof cost growth	4.6%/yr	BLS Producer Price Index, nonresidential roofing contractors, 2007–2025
Replacement cost	\$10.25/sq ft	San Antonio contractor price ranges
Reactive repair cost	3× planned × 1.3	Industry estimates; no primary study found. ×1.3 for leak damage in a school
Storm damage	0.38%/yr	Of roof cost, before insurance. FEMA National Risk Index expected building loss, roof-related hazards
PaxSeal fee	\$16,400/yr	PAX program estimate for this roof size. Your proposal will state the actual fee

Estimates, not a quote. Projections from San Antonio market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20. Storm damage is before insurance.